SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4316

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Case Study

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Code of Conduct:

I P.NIKITHA/192324226 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

P. Nikitha
Signature of the student:

Scenario: Web Development Standards Implementation

A startup is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET
B. POST
C. PUT
D. DELETE

Snippet 3: Navigation Elements

A.
B. <nav>
C. <section>
D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP
B. HTTP
C. SMTP
D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

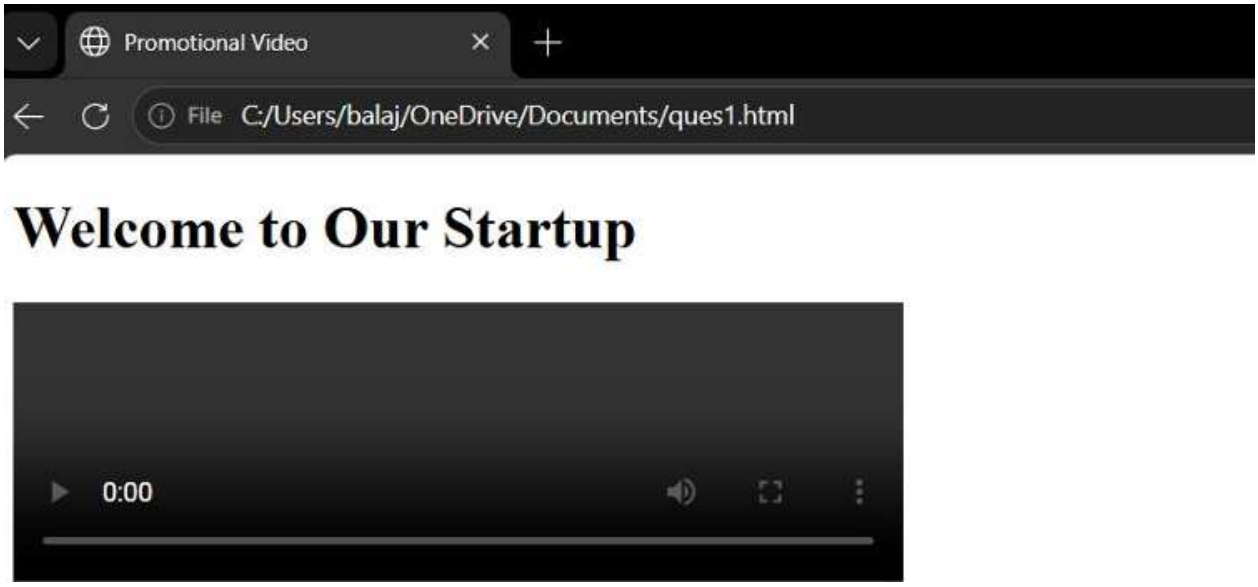
1. Correct HTML tag for video embedding

Correct answer: Option B — <video src="video.mp4" controls></video>

Explanation: The <video> tag is used to embed video content in an HTML webpage. The src attribute specifies the video file, and the controls attribute displays playback controls such as play, pause, and volume.

Example:

<video src="video.mp4" controls></video>



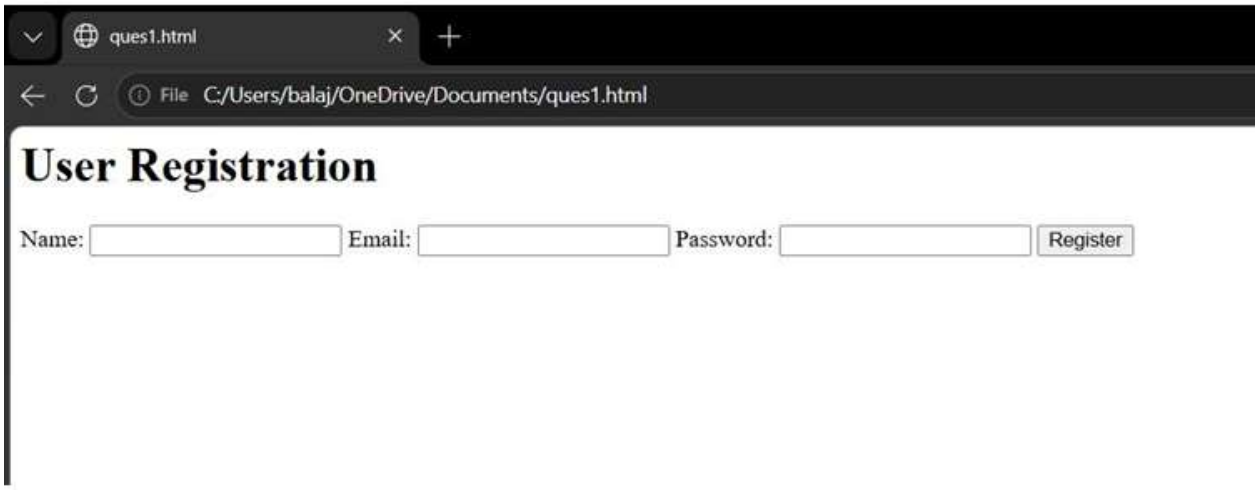
2. Appropriate HTTP method for form submission

Correct answer: Option B — POST

Explanation: The POST method sends form data to the server in the HTTP request body. It is suitable for registration forms because the submitted data is not placed directly in the URL, unlike GET.

Example:

```
<form action="/register" method="POST">
  <input type="text" name="username">
  <button type="submit">Register</button>
</form>
```



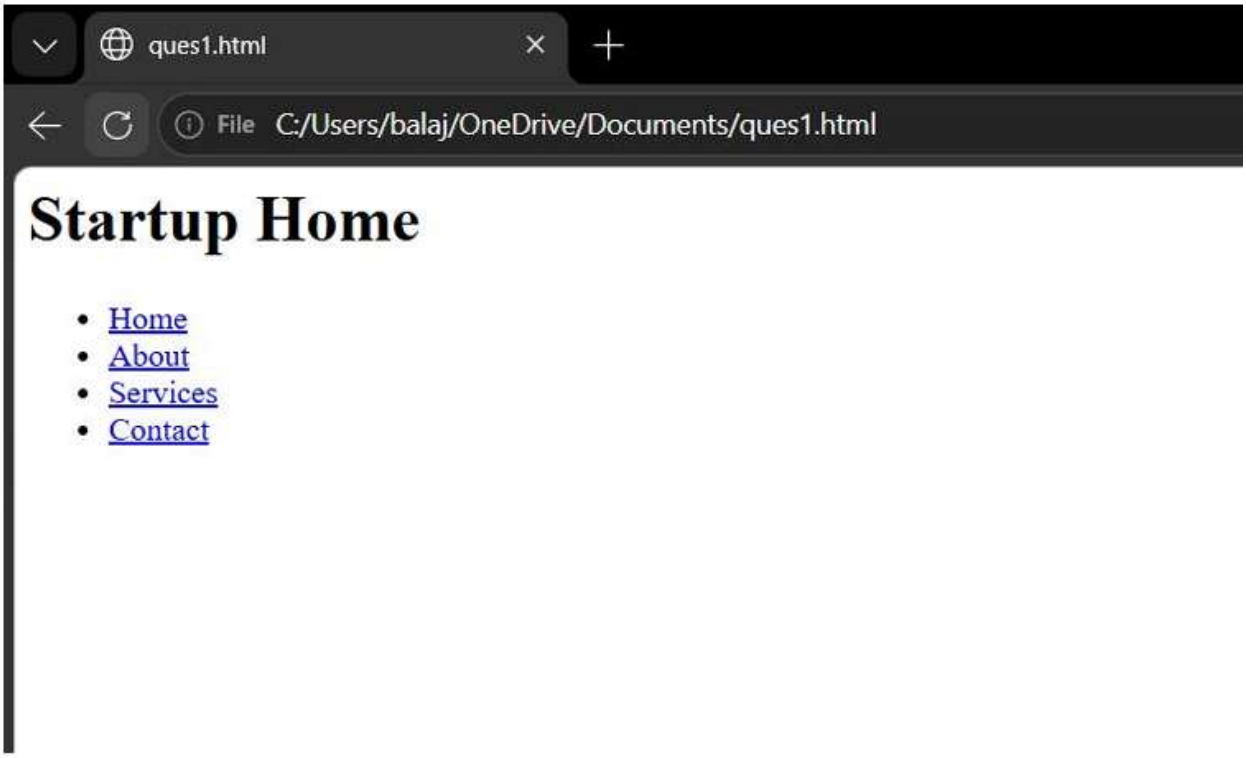
3. Proper semantic element for navigation

Correct answer: Option B — <nav>

Explanation: The <nav> element defines a section of a webpage containing major navigation links. It improves accessibility and helps browsers and assistive technologies understand the page structure.

Example:

```
<nav>
  <a href="index.html">Home</a>
  <a href="about.html">About</a>
  <a href="contact.html">Contact</a>
</nav>
```

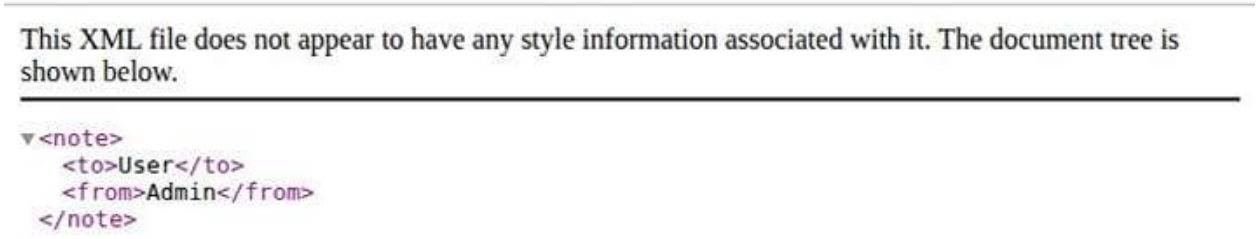


4. Correct XML syntax

Correct answer: Option B

<note>
  <to>User</to>
  <from>Admin</from>
</note>

Explanation: XML requires properly nested elements, matching opening and closing tags, and exactly one root element. Option B satisfies these rules.



5. Correct protocol for web communication

Correct answer: Option B — HTTP

Explanation: HTTP stands for Hypertext Transfer Protocol. It is used for communication between web clients, such as browsers, and web servers to request and deliver web resources.

HTTPS is the secure version of HTTP and is preferred for modern websites, especially those handling registration data.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand